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Dimensionality reduction for the analysis of brain oscillations

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ABSTRACT

Neuronal oscillations have been shown to be associated with perceptual, motor and cognitive brain operations. While complex spatio-temporal dynamics are a hallmark of neuronal oscillations, they also represent a formidable challenge for the proper extraction and quantification of oscillatory activity with non-invasive recording techniques such as EEG and MEG. In order to facilitate the study of neuronal oscillations we present a general-purpose pre-processing approach, which can be applied for a wide range of analyses including but not restricted to inverse modeling and multivariate single-trial classification. The idea is to use dimensionality reduction with spatio-spectral decomposition (SSD) instead of the commonly and almost exclusively used principal component analysis (PCA). The key advantage of SSD lies in selecting components explaining oscillations-related variance instead of just any variance as in the case of PCA. For the validation of SSD pre-processing we performed extensive simulations with different inverse modeling algorithms and signal-to-noise ratios. In all these simulations SSD invariably outperformed PCA often by a large margin. Moreover, using a database of multichannel EEG recordings from 80 subjects we show that pre-processing with SSD significantly increases the performance of single-trial classification of imagined movements, compared to the classification with PCA pre-processing or without any dimensionality reduction. Our simulations and analysis of real EEG experiments show that, while not being supervised, the SSD algorithm is capable of extracting components primarily relating to the signal of interest often using as little as 20% of the data variance, instead of > 90% variance as in case of PCA. Given its ease of use, absence of supervision, and capability to efficiently reduce the dimensionality of multivariate EEG/MEG data, we advocate the application of SSD pre-processing for the analysis of spontaneous and induced neuronal oscillations in normal subjects and patients.

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Introduction

The recent development of electrophysiological neuroimaging technology (e. g., EEG and MEG) has led to the possibility of recording brain activity using ever-denser sensor arrays. Often, for statistical, numerical and computational reasons, the number of obtained data dimensions is reduced for the following analyses. One of the most common approaches is to decrease the number of EEG/MEG channels by selecting for instance sensors covering certain parts of the head. Unfortunately, this approach has a big disadvantage as it assumes that cortical activity would produce EEG/MEG signals only in its immediate vicinity. However, due to the volume conduction property of the head, this is not the case and a given cortical current source can produce its maximal scalp electric potential or magnetic field far away from its origin.

A more adequate approach to dimensionality reduction is to consider a general lower-dimensional subspace of the data. The by far most-widely used dimensionality reduction pre-processing technique is principal component analysis (PCA), which is often used as a standard built-in feature in many program packages for the analysis of EEG/MEG data. PCA decomposes the data into orthogonal components maximally explaining the variance in the data. Importantly, the variance in the data is indiscriminative of its constituents, mixing together signal of interest and noise. Therefore, in case of EEG/MEG data, it is not always straight forward to attribute a given PCA component to signal or noise on the basis of its variance (that is, corresponding eigenvalue). Often, EEG and MEG data are characterized by strong artifactual components related to muscle contraction, eye blinking and other phenomena of extracerebral origin, while the brain activity of interest may be orders of magnitude weaker. Therefore, selecting components on the basis of explained variance might lead to the omission of important parts of the data (Jolliffe,

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1982). In some cases, PCA can be used to analyze EEG/MEG data; for instance, if the brain signals of interest are time-locked with respect to some known trigger, while all artifactual components are not. Here an averaging can be used to reduce the noise, making PCA pre-processing applicable, since in this case components with larger variance would indeed correspond to evoked responses. This, however, is not possible for oscillatory brain activity, which is typically not phase-synchronized with experimentally controlled events (Pfurtscheller and Lopez da Silva, 1999). Moreover, oscillatory dynamics are often studied without any reference to the external events, i. e., during resting state, where the interest is to understand dynamics of oscillations unfolding without perturbations (Hipp et al., 2012; Linkenkaer-Hansen et al., 2001; Nikouline et al., 2001; Nikulin and Brismar, 2005).

We argue that dimensionality reduction methods in neuroimaging in general and for oscillations in particular should be designed to clearly separate signal and noise subspaces. Importantly, here we are not necessarily interested in decomposition techniques which would separate individual brain sources. Even if a subspace of reduced dimensionality would contain a mixture of neuronal sources, this still would be an acceptable pre-processing result as we are more interested in avoiding for instance neuronal 1/f noise as much as possible. Moreover, since the criteria for categorizing data parts as signal or noise highly depend on the neurophysiological phenomena under study, dimensionality reduction should be tailored to the problem at hand. In the present study we are particularly interested in dimensionality reduction for the analysis of neuronal oscillations. The interest in oscillations stems from the fact that they are ubiquitous in the brain and are recordable with both invasive and non-invasive electrophysiology (e. g., with electro- and magnetoencephalography, electrocorticography, ECoG, and local field potential, LFP, recordings). Apart from their pervasive nature, neuronal oscillations have been implicated in practically all brain operations including perceptual, motor and cognitive functions (Buzsáki and Draguhn, 2004; Varela et al., 2001).

Nikulin et al. (2011) presented the spatio-spectral decomposition (SSD) algorithm for extracting rhythmic components, which are characterized by peaks in the power spectrum. They demonstrated that SSD is capable of successfully extracting individual neurophysiologically relevant brain rhythms with high signal-to-noise ratio (SNR). In this paper, we demonstrate that SSD can also serve as a versatile dimensionality reduction pre-processing tool for general analyses of oscillatory brain signals. To this end, we show that the use of SSD – as compared to using PCA – is the key to boosting the performance in two different and highly relevant application scenarios. Using realistically simulated EEG data, we demonstrate that SSD pre-processing better identifies the subspaces of oscillatory EEG activity than PCA, and thereby dramatically improves the accuracy with which the generators of these oscillations can be localized using EEG inverse calculations. Further, using real data, we show the benefit of performing SSD dimensionality reduction prior to common spatial patterns (CSP) analysis for the detection of different states of motor imagery in a brain–computer interfacing (BCI) context.

The paper is structured as follows. In the Dimensionality reduction section, we introduce PCA and SSD within the context of linear dimensionality reduction. The Localization of simulated oscillatory EEG sources section describes an extensive numerical simulation study, while the Classification of motor imagery conditions section presents results obtained on real EEG data in a BCI context. Results of both studies are shown in the Results section, and implications for future studies on neural oscillations are discussed in the Discussion section.

Methods

In the following, we introduce two general ways in which linear data decompositions can be used to denoise data. Besides the classical 'dimensionality reduction' setting, in which the denoised data are represented in a lower-dimensional coordinate system, we also deal with 'low-rank factorizations' of the data, in which the original coordinate system is maintained, the data, however, are constrained to lie in a lower-dimensional subspace. Moreover, we introduce principal component analysis and spatio-spectral decomposition as dimensionality reduction techniques, and the amount of variance explained as an important quantity to characterize such techniques.

Dimensionality reduction

In the following, we denote the multivariate neuronal measurement acquired using EEG, MEG, ECoG or LFP at time t by $\mathbf{x}(t) \in \mathbb{R}^M$. In linear *dimensionality reduction*, the measurements are projected onto a lower dimensional space by means of a projection matrix $\mathbf{W} \in \mathbb{R}^{M \times K}$, $K < M$:

$$\tilde{\mathbf{x}}(t) = \mathbf{W}^\top \mathbf{x}(t) \quad . \tag{1}$$

Subsequent analyses are then performed on the dimensionality-reduced data $\tilde{\mathbf{x}}(t)$.

The classical advantages of linear dimensionality reduction are:

- **Computational efficiency**: The computational load of subsequent analyses is reduced by working on lower-dimensional data.
- **Regularization**: If the number of parameters of subsequent multivariate methods grows with the number of data dimensions, overfitting can occur for high-dimensional datasets, reducing the model's ability to make predictions for new data points. Performing dimensionality reduction prior to such analyses is a classic form of regularization effectively reducing the degrees of freedom of subsequent methods and thereby preventing overfitting.
- **Numerical stabilization**: Dimensionality reduction can stabilize numerical computations in subsequent analyses by decreasing the redundancy/collinearity in the data. Many methods require the data matrices to be linearly independent and well-conditioned (e. g. for obtaining a stable estimate of the inverse of the data covariance matrix). In high-dimensional datasets, both requirements are often compromised, but can be reestablished using appropriate dimensionality reduction.
- **Denoising**: Dimensionality reduction can also provide a decomposition of the data such that all the signals of interest for subsequent analyses are included in the retained dimensions, while dimensions containing only or mainly noise (i. e., signals-of-no-interest) are discarded. Clearly, any signal parts being removed are likely to have a negative effect on subsequent analyses, as do noise parts being retained.

Low-rank factorization

In some situations, it is necessary to perform subsequent analyses in the original input space, e. g., if one wishes to interpret topographical maps resulting from these analyses. While the first three classical benefits of dimensionality reduction are of less importance here, one may still desire to achieve noise (e. g., artifact) reduction. To this end, a 'denoised' *rank-K factorization* of $\mathbf{x}$ can be obtained by means of the transformation

$$\tilde{\tilde{\mathbf{x}}}(t) = \mathbf{A}\mathbf{W}^\top \mathbf{x}(t) \quad . \tag{2}$$

Here, $\mathbf{A} \in \mathbb{R}^{M \times K}$ is the *activation pattern* (forward projection) matrix corresponding to the *extraction filter* (backward projection) matrix $\mathbf{W}$ (see Haufe et al., 2014). Most dimensionality reduction techniques including PCA and SSD provide both $\mathbf{W}$ and $\mathbf{A}$ (see also Low-rank factorization and Localization of simulated oscillatory EEG sourcessections). If only $\mathbf{W}$ is provided, $\mathbf{A}$ can be obtained by

$$\mathbf{A} = \Sigma_{\mathbf{x}} \mathbf{W} \Sigma_{\tilde{\mathbf{x}}}^{-1} \quad , \tag{3}$$

where $\Sigma_{\mathbf{x}}$ and $\Sigma_{\widetilde{\mathbf{x}}}$ are the empirical covariance matrices of $\mathbf{x}(t)$ and $\widetilde{\mathbf{x}}(t)$, respectively (Haufe et al., 2014). Note that if the extraction filters were optimized on bandpass filtered data, the covariance matrices used in Eq. (3) must also be obtained on bandpass filtered data.

In the remainder of this manuscript, we will make the distinction between 'dimensionality reduction' and 'low-rank factorization' only when necessary, and otherwise use the term 'dimensionality reduction' for both approaches.

Principal components analysis

Principal component analysis rotates the data into directions of maximal variance. Assuming centered (zero-mean) data, the first PCA component is obtained by

$$\mathbf{w}_1 = \arg\max_{\mathbf{w}} \frac{\mathbf{w}^\top \Sigma_{\mathbf{x}} \mathbf{w}}{\mathbf{w}^\top \mathbf{w}} \quad . \tag{4}$$

Subsequent components $\mathbf{w}_2,\ldots, \mathbf{w}_M$ are obtained by subtracting the previous components via

$$\mathbf{x}_K(t) \leftarrow \mathbf{x}_{K-1}(t) - \mathbf{w}_{K-1}\mathbf{w}_{K-1}^\top \mathbf{x}_{K-1}(t) \tag{5}$$

and applying Eq. (4) recursively. The PCA filter matrix $\mathbf{W} = [\mathbf{w}_1,\ldots, \mathbf{w}_M]$ is equivalently obtained through an eigenvalue decomposition of the data covariance matrix:

$$\Sigma_{\mathbf{x}} = \mathbf{W}\mathbf{D}\mathbf{W}^\top \quad , \tag{6}$$

where $\mathbf{D}$ is a diagonal matrix containg eigenvalues (the variances in each PCA direction) of $\Sigma_{\mathbf{x}}$ on the diagonal. While the transformation from input channels to PCA components is given by $\widetilde{\mathrm{x}}(t) = \mathbf{W}^\top \mathbf{x}(t)$, the inverse transformation is given by $\mathbf{x}(t) = \mathbf{A}\widetilde{\mathbf{x}}(t)$ with $\mathbf{A} = \mathbf{W}^{-\top}$, where $\mathbf{W}^{-\top}$ denotes the inverse of $\mathbf{W}^\top$. Since for PCA $\mathbf{W}$ can be proven to be orthogonal, we have that $\mathbf{W}^\top = \mathbf{W}^{-1}$, and thus $\mathbf{A} = \mathbf{W}$. Let w. l. o. g. the entries of $\mathbf{D}$ be sorted in decreasing order. The number of PCA components needed to explain p percent of the variance in the data can be determined by evaluating

$$p > \frac{100 \sum_{i=1}^{K} d_i}{\sum_{i=1}^{M} d_i} \tag{7}$$

for $K = 1, \ldots, M$. The smallest K, for which Ineq. (7) holds, is chosen.

Spatio-spectral decomposition

The purpose of SSD (Nikulin et al., 2011) is to extract brain oscillations with high SNR[2]. Defining the SNR as the ratio of the power in the frequency band of interest and the power in the left and right neighboring (flanking) frequency bands, it is maximized by

$$\mathbf{w}_1 = \arg\max_{\mathbf{w}} \frac{\mathbf{w}^\top \Sigma_{\mathrm{sig}} \mathbf{w}}{\mathbf{w}^\top \Sigma_{\mathrm{noise}} \mathbf{w}} \quad , \tag{8}$$

where Σ_{sig} is the covariance of the data filtered in the frequency band of interest, while Σ_{noise} is the covariance of the data filtered in the sidebands. Practically, filtering in the left and right side bands can be implemented by applying a bandpass filter covering both side bands followed by a bandstop filter cutting out the band-of-interest. The entire SSD filter matrix can be equivalently obtained by means of the generalized eigenvalue decomposition of Σ_{sig} and Σ_{noise}

$$\Sigma_{\mathrm{sig}} \mathbf{W} = \Sigma_{\mathrm{noise}} \mathbf{W}\mathbf{D} \quad . \tag{9}$$

[2] A Matlab implementation of SSD can be downloaded using the following URL: http://www.user.tu-berlin.de/sdaehne/code/ssd/.

Previously, SSD was shown to extract individual oscillatory sources in EEG better and considerably faster than corresponding decompositions on the basis of independent components analysis (Nikulin et al., 2011). For selecting the number of components we propose a heuristic based on evaluating the ratio of the power in the passband and the power in the sidebands (the generalized eigenvalues of Eq. (9)) for each SSD component. The reasoning here is that for true oscillatory components, this ratio should stand out against all other components. Thus, we can use a standard measure for detecting 'outliers' in data such as the interquartile range (IQR, Upton and Cook, 1996). Specifically, we define a threshold

$$\mathrm{thr} = Q_{75\%} + q(Q_{75\%} - Q_{25\%}) \quad , \tag{10}$$

where $Q_{25\%}$ and $Q_{75\%}$ are the first and third quartiles of the power ratios of all SSD components and q is a nonnegative constant. For a given q, we select all SSD components for which the threshold is exceeded.

Calculation of explained variance

Assuming zero-mean data, the variance in the measurements explained by a given number of K (e. g., PCA or SSD) components is given by $\mathrm{EV} = \|\widetilde{\widetilde{\mathbf{X}}}\|_F^2 / \|\mathbf{X}\|_F^2$, where $\mathbf{X} = [\mathbf{x}(1), \ldots, \mathbf{x}(T)] \in \mathbb{R}^{M \times T}$ are the original measurements, $\widetilde{\widetilde{\mathbf{X}}} = \left[\widetilde{\widetilde{\mathbf{x}}}(1), \ldots, \widetilde{\widetilde{\mathbf{x}}}(T)\right] \in \mathbb{R}^{M \times T}$ is their rank-K factorization, and $\|\mathbf{X}\|_F^2 = \sum_{ij} \mathbf{X}_{ij}^2$ is the sum of the squared entries of a matrix. This definition is formally equivalent to the definition used in Borisov et al. (2006) in the context of independent component analysis.

Localization of simulated oscillatory EEG sources

Here we describe a large-scale simulation study intended to benchmark PCA and SSD with respect to their ability to recover brain oscillations. In addition, the study has the purpose to quantify the effectiveness of the subspace separation provided by PCA and SSD in a widely considered application scenario, namely EEG inverse source reconstruction. It is well-known that, due to volume conduction in the head, sensor-space EEG measurements cannot be interpreted in terms of the physical locations of the underlying brain sources unless the so-called inverse problem is solved. Technically speaking, the inverse problem consists of decomposing the data into

$$\mathbf{x}(t) = \mathbf{L}\mathbf{j}(t) + \epsilon(t) \quad , \tag{11}$$

thereby dividing source mixing due to volume conduction in the head as modeled by the so-called *lead field* $\mathbf{L}$ from the actual brain source electrical activity $\mathbf{j}(t)$, where $\epsilon\ (t)$ is potential measurement noise. To link the estimated source activity $\mathbf{j}(t)$ to brain locations, $\mathbf{L}$ is typically a predetermined $M \times 3V$ matrix containing the theoretical spread of $V \gg M$ unit-length dipoles located on a regular grid in the brain (or along the cortical manifold) and oriented in the x- y- and z- directions. Solving Eq. (11) for $\mathbf{j}(t)$ is mathematically ill-posed and can be performed only if additional constraints on the solution are imposed. In addition to that, noise contributions such as artifacts, measurement noise and brain processes of no interest can hamper the correct localization of the brain processes under study. It is therefore desirable to suppress noise sources as much as possible prior to source localization. While averaging can be used for phase-locked signals such as event-related potentials (ERPs), the same cannot be done for neural oscillations. Working on the low-rank 'denoised' signal (Eq. (2)), however, bears the potential to improve the localization of brain oscillations, which is tested here.

Data generation

We simulated pseudo-EEG data as follows. In each experiment, K_s oscillatory signal components $\mathbf{s}_s(t) \in \mathbb{R}^{K_s}$ were generated by filtering independent standard normal distributed noise in the frequency band of the alpha rhythm (8–13 Hz) using a 5-th order causal Butterworth filter. The sampling frequency of the data was 200 Hz, and $T = 60\,000$ samples were generated, amounting to 5 min of data. The K_s components were mapped to $M = 118$ EEG sensors (extended international 10–20 system) via

$$\mathbf{x}_s(t) = \mathbf{A}_s \mathbf{s}_s(t) \ , \tag{12}$$

where $\mathbf{A}_s \in \mathbb{R}^{M \times K_s}$ is the theoretical field spread of K_s equivalent current dipoles, which were randomly placed in the brain volume and had random spatial orientation. The forward calculations were carried out in the 'ICBM 2009b Nonlinear Symmetric' head model, a nonlinear average of the anatomic magnetic resonance (MR) images of 152 adults (Fonov et al., 2011) using semi-analytic expansions of the electric lead fields (Nolte and Dassios, 2005).

The signal-related pseudo-EEG data were superimposed with the activity of $K_n = 500$ simulated 'brain noise' components $s_n(t) \in \mathbb{R}^{K_n}$. To reflect typical EEG recordings, these components were generated independently as pink noise processes characterized by a 1/f-shaped power spectrum. Similar to the signal components, the simulated noise time courses $s_n(t)$ were mapped to the 118 EEG sensors via

$$\mathbf{x}_n(t) = \mathbf{A}_n \mathbf{s}_n(t) \tag{13}$$

using a matrix $\mathbf{A}_n \in \mathbb{R}^{M \times K_n}$ containing the theoretical field spread of 500 dipoles with random locations and orientations.

We constructed an alpha-band filtered version of the noise using the same causal Butterworth filter described above. Let the filtered noise be denoted by $\mathbf{x}_{fn}(t)$ and be subsumed in the $M \times T$ matrix $\mathbf{X}_{fn}$. Similarly, let the signal- and noise-related EEG data $\mathbf{x}_s(t)$ and $\mathbf{x}_n(t)$ be subsumed in the matrices $\mathbf{X}_s$ and $\mathbf{X}_n$. We define the signal-to-noise ratio (SNR) as the ratio of the power of the signal- and noise-related EEG time series *within the alpha-band*. Using this definition, the pseudo-EEG for a given SNR λ was generated by

$$\mathbf{x}(t) = \lambda \frac{\mathbf{x}_s(t)}{\|\mathbf{X}_s\|_{\mathcal{F}}} + \frac{\mathbf{x}_n(t)}{\|\mathbf{X}_{fn}\|_{\mathcal{F}}} \ . \tag{14}$$

We conducted $N = 100$ experiments for each combination of $K_s \in \{1, 3, 5\}$ and $\lambda \in \{1/3, 1/10\}$.

Note that the data simulated here correspond to a 'resting state' measurement; therefore, no additional 'control' condition was simulated. We deliberatly chose a resting state setting, since it is the most challenging yet realistic (since most brain rhythms are strongest during rest) setting for the analysis of brain oscillations. In a task paradigm, better estimates may be expected when contrasting results obtained for control (often the resting state itself) and task conditions, or by using the control condition to obtain a realistic noise estimate (e. g., for LCMV source localization).

Low-rank factorization

In each experiment, we generated low-rank factorizations of the original data using PCA and SSD in order to obtain 'denoised' signals. The reason for using low-rank factorizations according to Eq. (2) rather than dimensionality reduction according to Eq. (1) are outlined in the Discussion section. For PCA, we filtered the pseudo-EEG observations $\mathbf{x}(t)$ in the alpha-band. In order to cover a range of explained variance, we used PCA components accounting for either 90% or 99% of the variance in the observed data. To obtain estimates of the peak performances of PCA and SSD, we moreover applied both methods using $K = K_s$. That is, we set the number of selected components to the true number of simulated brain oscillations. The two resulting methods are called PCA KTRUE and SSD KTRUE, respectively. Finally, to obtain an estimate of the performance maximally achievable by any method, we also benchmarked projections to the 'true' subspace of the underlying simulated oscillations using the actual $M \times K_s$ matrix $\mathbf{A}_s$ that was used to project the activity of the simulated dipoles to the EEG sensors in the realistic head model. To remove all data parts orthogonal to the columns spanned by $\mathbf{A}_s$, we used the following formula:

$$\tilde{\tilde{\mathbf{x}}}(t) = \mathrm{orth}(\mathbf{A}_s)\mathrm{orth}(\mathbf{A}_s)^\top \mathbf{x}(t) \ , \tag{15}$$

where $\mathrm{orth}(\mathbf{A}_s)$ is an orthogonal $M \times K_s$ basis for the span of $\mathbf{A}_s$. The resulting method is called ATRUE.

Note that PCA KTRUE, SSD KTRUE and ATRUE use information that is not available in practice, and therefore only serve as baselines for benchmarking PCA 99, PCA 90, SSD AUTO as well as the absence of any dimensionality reduction, denoted by NO DIMRED.

Performance measures: subspace angle and single component reconstruction accuracy

We measured the performance, with which PCA and SSD recovered the simulated oscillations, in two ways. First, the angle between the subspaces spanned by the columns of the estimated activation pattern matrix $\mathbf{A}$ and the true matrix $\mathbf{A}_s$ was evaluated. This angle ranges from 0° for perfectly aligned subspaces to 90° for completely non-overlapping (perpendicular) subspaces. Second, we measured the reconstruction of single oscillatory sources. To this end, we optimally matched the columns of $\mathbf{A}$ to those of $\mathbf{A}_s$ using the Kuhn-Munkres Algorithm (Tichavský and Koldovský, 2004). The measure of pairwise similarity between columns of $\mathbf{A}$ and $\mathbf{A}_s$ used within the algorithm was the goodness-of-fit (GOF) achieved by regressing one column onto the other using linear ordinary least-squares (OLS) regression, as in (Haufe et al., 2010). The GOF across all columns was assessed as a measure of average reconstruction of the individual oscillatory sources. This quantity ranged from 0 for totally uncorrelated simulated and estimated field patterns to 1 for perfect recovery of each individual field pattern.

The goodness-of-fit criterion is based on a one-to-one matching between the columns of $\mathbf{A}$ and $\mathbf{A}_s$; therefore, it is required that $K = K_s$. Similarly, angles can only be meaningfully compared between subspaces of the same dimensionality, since they generally decrease with growing dimensionality until eventually vanishing for $K = M$. Therefore, both performance measures were evaluated only for PCA KTRUE and SSD KTRUE.

Performance measures: source localization accuracy

In addition to measuring the mere reconstruction of the signal components – and thereby the separation of signal and noise components – we also quantified the benefit of this subspace separation for the localization of the generators of the simulated oscillations. To this end, we applied various inverse source reconstruction techniques to the bandpass-filtered original pseudo-EEG time series (NO DIMRED), as well as to the various 'denoised' variants outlined above. For PCA 99, PCA 90, SSD AUTO, PCA KTRUE and SSD KTRUE, Eq. (2) was used to construct the denoised signal, while for ATRUE, Eq. (15) was used. The inverse methods employed were the linearly-constrained minimum variance (LCMV) beamformer (Van Veen et al., 1997), exact low resolution brain electromagnetic tomography (eLORETA, Pascual-Marqui, 2007) and sparse basis field expansions (S-FLEX, Haufe et al., 2009; Haufe et al., 2011). Since the simulation scenario did not comprise measurement noise (c. f., Eq. (14)), the simulated pseudo-EEG could be perfectly explained by electrical sources within the brain. Therefore, all inverse methods were run with only a small amount of regularization. For LCMV, a multiple of the identity matrix was added to the data covariance matrix. The factor was determined using automatic shrinkage (Blankertz et al., 2011; Ledoit and Wolf, 2004). The regularization constant in eLORETA was set to $\alpha = \mathrm{tr}(\mathbf{A}_l \mathbf{W}^{-1} \mathbf{A}_l^\top)/(10^3 M)$ in each iteration, where $\mathbf{W}$ is the eLORETA weight matrix. Lastly, S-FLEX was adjusted such that estimated source activity accounted for 99.75%

of the variance in the pseudo-EEG data. The reconstruction was carried out in a source space of $V = 2\ 113$ voxels using the same head model in which the simulated data had been generated. Both LCMV and eLORETA are computationally comparably inexpensive procedures, which is explained by the fact that LCMV does not model dependencies between voxels and is thus applied separately to each voxel, while eLORETA does not take the temporal structure of the data into account and is applied separately to each sample. S-FLEX, on the other hand, is a nonlinear method modeling both spatial and temporal dependencies using sparsity constraints. As a consequence, S-FLEX localizes entire time-series using a single large-scale iterative numerical optimization, which is time-consuming. To reduce the computational load here, S-FLEX was applied only to reduced data composed of 100 randomly selected samples.

To measure the quality of the inverse source reconstructions, the total energy (power) of the source activity was calculated for each voxel yielding a V-dimensional power-distribution for each method. This distribution was normalized to sum to one. For LCMV, the power in each voxel was additionally normalized by the power of white sensor noise to avoid an otherwise present bias of this method towards central locations. The normalized power is termed 'neural activity index' and is defined in Eq. (27) of (Van Veen et al., 1997). The power distribution was also calculated for the true simulated source distribution consisting of K_s dipoles.

In order to compare the true power distribution with the estimated power distribution, we applied a metric called the *earth mover's distance* (EMD, Rubner et al., 2000; Haufe et al., 2008). Intuitively speaking, the EMD quantifies the difference between two distributions by measuring the costs required for transforming one distribution into the other. In the EMD metaphor, one distribution is seen as a mass of earth spread in space, while the other distribution is seen as a collection of holes in that same space. The EMD measures the least amount of work needed to fill the holes with earth, where a unit of work corresponds to transporting a unit of earth by a unit of distance. The EMD allows for a meaningful comparison of distributions of different types, e. g., very sparse versus very smooth, or distributions with non-overlapping support. For the formal definition and a thorough discussion of the properties of the EMD see (Rubner et al., 2000). We computed the EMD using the implementation of Pele and Werman, 2009 and the Euclidean distance in 3D voxel space as the ground metric.

Classification of motor imagery conditions

In this section we describe the use of SSD as a dimensionality reduction method in the context of a brain–computer interface (BCI) experiment. We consider data from a BCI study with $N = 80$ subjects, in which the subjects had to imagine movements of their left/right hands, or their feet. The data were originally collected for a large-scale screening study on BCI aptitude (Blankertz et al., 2010), which was approved by the Ethical Review Board of the Medical Faculty of the University of Tübingen.

Imagined movements of body parts lead to event-related (de-) synchronization (ERD/ERS) of spatially distinct generators of sensorimotor rhythmic (SMR) activity, namely those linked to the imagined body parts. These localized ERD phenomena, i.e. the modulations of specific brain rhythms, are the neuronal basis for motor imagery (MI) based brain–computer interfaces. In the following paragraphs we describe the common analysis steps for single-trial classification in MI-BCI.

BCI paradigm

Here we use the data from a calibration session only. In this session, the subjects were instructed to perform motor imagery of their left hand, right hand, or feet in randomized order after presentation of a respective cue. A total of 225 trials (75 per class) of 5 s length were recorded. For each pair of classes, a best-discriminating frequency band in the SMR range (5–30 Hz) and time interval relative to the cue was determined heuristically (Blankertz et al., 2008). On the basis of the calibration data, the best-discriminating motor imagery class combination (e.g. imagination of left hand versus foot movement) was selected and used in the feedback part of the experiment. In the present analysis, we only used trials of the two best-discriminable MI classes from the calibration data, which amounts to a total of 150 trials.

Data acquisition and artifact rejection

EEG signals were recorded using a Fast'n Easy Cap (EasyCap GmbH) with 119 Ag/AgCl electrodes placed at symmetrical positions based on the international 10–20 system. The channel setup included horizontal and vertical electrooculographic (EOG) channels, which were not used in the present analysis. EEG date were recorded and stored at 1000 Hz sampling rate, with a band-pass filter of 0.5 Hz to 200 Hz. For the offline analysis, the data were low-pass filtered with a cutoff frequency at 45 Hz and downsampled to 100 Hz sampling rate.

Trials contaminated with artifacts were identified and removed with an automatic procedure, based on a variance criterion. To this end, the EEG data were band-pass filtered between 5 Hz and 45 Hz, and the variance was computed for each channel and trial separately and then averaged across channels. This yielded a single *global variance* value per trial. Trials with a global variance larger than a subject-specific threshold were discarded. The threshold $\mathrm{thr}_{\mathrm{arti}}$ for artifact trial rejection was set according to

$$\mathrm{thr}_{\mathrm{arti}} = Q_{95\%} + 3 \cdot \left(Q_{95\%} - Q_{5\%}\right) \ , \tag{16}$$

where $Q_{5\%}$ and $Q_{95\%}$ denote the 5 and 95 percentiles of the subject-specific global variance distribution.

Common spatial patterns (CSP)

In MI-BCI, spatial filters are commonly applied to the raw data in order to reduce the effects of volume conduction and to thereby increase the signal-to-noise ratio. This greatly improves the detection of changes in SMR activity and thus enables single-trial classification of motor-related oscillatory activity. A particularly popular spatial filtering technique in the context of MI-BCIs is the so-called common spatial patterns (CSP, Blankertz et al., 2008; Koles, 1991) algorithm.

Given bandpass-filtered data from a two-class BCI paradigm, CSP optimizes spatial filters that maximize the variance of the spatially filtered signal for one class, while minimizing the variance for the respective other class. The variance of the bandpass-filtered signal here serves as an approximation of the band-power (because band-pass filtering leads to zero mean signals and thus the following squaring effectively leads to an estimate of the variance/power) and thereby represents as a measure of ERD/ERS. One formulation of the CSP objective function is given by

$$\mathbf{w} = \arg\max_{\mathbf{w}} \frac{\mathbf{w}^\top (\Sigma_1 - \Sigma_2)\mathbf{w}}{\mathbf{w}^\top (\Sigma_1 + \Sigma_2)\mathbf{w}} \ , \tag{17}$$

where Σ_1 denotes the covariance matrix of all trials of class 1 (e. g., left hand imagery), while Σ_2 denotes the covariance matrix of class 2 (e. g., right hand imagery). The entire matrix of CSP filters can be obtained by means of the generalized eigenvalue decomposition of the matrices $(\Sigma_1 - \Sigma_2)$ and $(\Sigma_1 + \Sigma_2)$,

$$(\Sigma_1 - \Sigma_2)\mathbf{W} = (\Sigma_1 + \Sigma_2)\mathbf{W}\mathbf{D} \ , \tag{18}$$

where the diagonal entries of $\mathbf{D}$ are the generalized eigenvalues of both matrices, containing the ratios of the variances of the trials of both classes when projected onto the CSP filters. Interestingly, the columns of $\mathbf{W}$ contain not only the spatial filter that maximizes this ratio and thus Eq. (17), but also the one that minimizes it. Let the columns of $\mathbf{W}$ be ordered according to the corresponding generalized eigenvalues in the

diagonal of **D**, in decreasing order. Then the first and last column of **W** are the projection vectors that maximize the variance ratio between the two classes. In practice, not only the first and the last eigenvector are chosen but two or three from each side of the corresponding eigenvalue spectrum.

Classification

The single-trial decision between MI classes is commonly carried out by a linear classifier such as linear discriminant analysis (LDA).

LDA optimizes weighting coefficients $\{\beta_j\}_{j=0}^{J}$ which integrate the information from J CSP components and estimates a label $\hat{y}(e) \in \{-1, +1\}$ for a given trial with index e according to

$$\hat{y}(e) = sign\left(\sum_{j=1}^{J} \beta_j log\left(\mathbf{w}_j^{\top} \Sigma_e \mathbf{w}_j\right) + \beta_0\right) \quad , \tag{19}$$

where Σ_e denotes the covariance matrix of the band-passed filtered signal in trial e and $\mathbf{w}_j$ is one of J CSP components. The weighting coefficients $\{\beta_j\}_{j=0}^{J}$ are optimized to minimize the error between the true class labels y and the estimated labels $\hat{y}$ on a number of training trials.

A test trial is classified by evaluating Eq. (19), i. e., by first projecting the data onto the CSP components, then computing the log-variance for each component and finally combining the resulting log-variance values using the LDA weighting coefficients. The sign of the resulting value then indicates the estimated class label.

Parameters: frequency bands and time intervals

Subject-specific parameters such as the class-discriminative frequency band and within-epoch time intervals were taken from the original study on BCI aptitude (Blankertz et al., 2010), during which these parameters were determined by the experimenter manually and with the help of heuristics on the basis of the calibration data. Frequency bands were within the limits of the alpha and beta range, i. e., with low cutoff frequency being 8 Hz and high cutoff frequency being 30 Hz. Time intervals were limited to the range from 500 ms to 4500 ms after the indication of the cue on the screen.

Dimensionality reduction

In order to potentially benefit from the advantages of linear dimensionality reduction outlined in the Dimensionality reduction section, we projected the data into lower-dimensional spaces using PCA and SSD according to Eq. (1). For PCA, the data of each subject were band-pass filtered in the class-discriminative band (from here on referred to as the signal band), and trials were reduced to the class-discriminative intervals. PCA was performed via an eigen-decomposition of the covariance matrix of the resulting data. The PCA components were ordered according to their variance (i. e., the eigenvalue in Eq. (6)) from highest to lowest.

For SSD, two covariance matrices are necessary: one that captures the covariance of the *signal* and one for the *noise*. The signal covariance matrix was the same as was used for PCA. For the covariance of the side bands, i.e. the noise part, we employed a combination of band-pass and band-stop filtering. Firstly, the data were band-pass filtered with passband boundaries that were 3 Hz wider to either side of the signal pass-band. Thereafter, the data were bandstop-filtered with stopband boundary frequencies that were only 1 Hz wider than the signal passband. This procedure effectively bandpass-filtered the data in two sidebands, each 2 Hz wide, and separated by 1 Hz below and above the signal passband. The noise covariance matrix was calculated in the predefined time-interval. SSD components were calculated as the result of the generalized eigen-decomposition (see Eq. (9)), and the resulting components were ordered according to their corresponding generalized eigenvalues from highest to lowest.

The dimensionality of the bandpass-filtered data was reduced using PCA/SSD, before the data were subjected to CSP analysis. The minimum number of PCA/SSD components was set to six, because we always used three components from either side of the CSP eigenvalue-spectrum, corresponding to the three highest and three lowest CSP eigenvalues (see Eq. (18)).

Since CSP and LDA make use of the class label information, they were applied using 5-fold chronological cross-validation (Lemm et al., 2011): the trials were split into five equally sized partitions (also called folds), of which four served as training data to optimize CSP components and LDA coefficients, while the remaining fifth partition was used to test the classification performance. This was repeated five times such that each partition became the test data once, and the resulting classification accuracies were averaged.

Performance measure: classification accuracy

The baseline for classification performance is given by applying CSP without any dimensionality reduction. The outlined cross-validated testing procedure was repeated for increasing numbers of PCA and SSD components, starting with six components and adding more components according to the ranking imposed by their corresponding (generalized) eigenvalues. The performance measure in our experiment was the classification performance averaged over cross-validation folds.

Performance was assessed also for several rules for PCA/SSD component selection. Similar to our simulations, we apply the rules of 90% and 99% explained variance to select the number of PCA components, denoting the corresponding methods with PCA 90 and PCA 99, respectively. For SSD, we use the heuristic outlined in the Low-rank factorization section with $q = 0.5$, and denote the corresponding method with SSD AUTO. Note that we required a minimum of six PCA/SSD components as input to CSP. Thus the actual number of components used was the maximum of six and the number suggested by the PCA/SSD rule.

Additionally, we tested an SSD version in which the number of selected components was fixed to 14, denoted by SSD 14. As will be shown in the Results section, this number of components yielded best performance on average over all subjects and thus served as an upper bound for the heuristic selection rules.

In order to assess robustness and practical relevance of the results, the experiment was conducted separately using three different subsets of the originally recorded 119 EEG electrode montage. The symmetrically placed subsets included 33, 57, and 116 channels, where the layout of each subset montage is shown as insets in Figs. 4 and 5.

Finally, we investigated the proportion of total signal variance explained by subsets of K PCA or SSD components.

Results

Localization of simulated oscillatory EEG sources

Subspace angle and single component reconstruction accuracy

Fig. 1 depicts the performance of PCA and SSD according to various measures as box plots. For each box, the central mark is the median, the edges of the box are the 25th and 75th percentiles, and the whiskers extend to the most extreme datapoints. For each box, a comparison interval is highlighted as a notch, where two medians are significantly different at the 5% level, if the notches do not overlap.

The top row of Fig. 1 depicts the angle between the subspaces spanned by the true and estimated activation pattern matrices $\mathbf{A}_s$ and $\mathbf{A}$ for PCA KTRUE and SSD KTRUE. It can be seen that SSD KTRUE outperforms PCA KTRUE with significantly lower subspace angles in all but the two most challenging scenarios $\lambda = 1/10$ with $K_s \geq 3$, where both methods are on par.

The middle row of Fig. 1 depicts the error with which each single (1-dimensional) simulated brain oscillation is recovered by PCA KTRUE and SSD KTRUE. Here, SSD KTRUE significantly outperforms PCA KTRUE

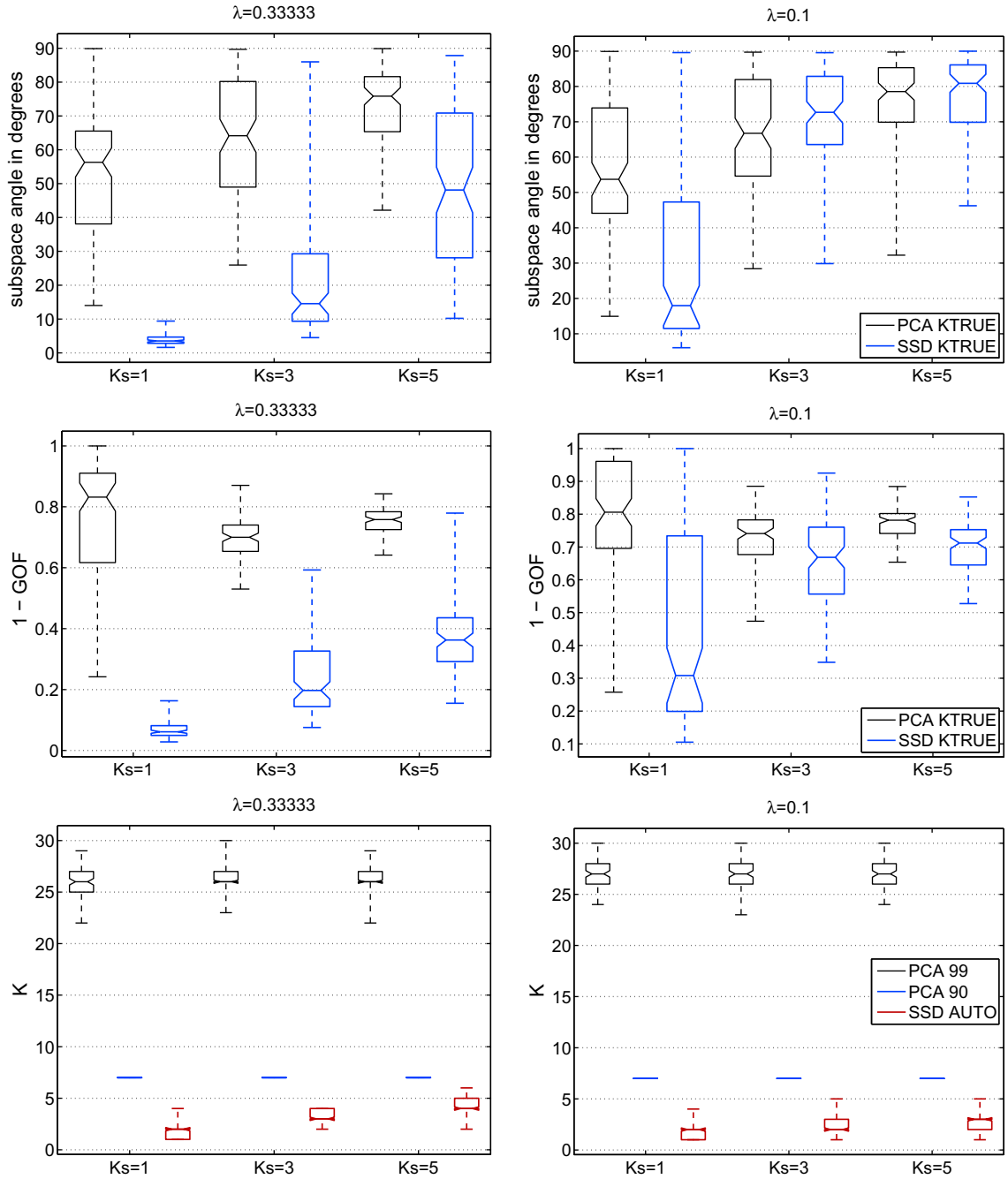

Fig. 1. Ability of principal component analysis (PCA) and spatio-spectral decomposition (SSD) to reconstruct brain oscillations from simulated scalp electroencephalographic (EEG) signals for two signal-to-noise ratios ($\lambda \in \{1/3, 1/10\}$) and three numbers of simulated oscillatory components ($K_s \in \{1, 3, 5\}$). Top row: angle between the subspaces spanned by the true and estimated activation pattern matrices $\mathbf{A}_s$ and $\mathbf{A}$. To allow a fair comparison of the methods, the number of components was fixed to the true number $K = K_s$. The resulting methods are called PCA KTRUE and SSD KTRUE. Middle row: error with which the activation patterns (scalp potential maps) of individual simulated oscillatory brain components are recovered. In order to be able to match estimated and true patterns, the number of estimated components was again set to $K = K_s$. Bottom row: number of estimated components if heuristics are used. For PCA, the common 90% and 99% rules were used. The resulting methods are called PCA 90 and PCA 99, respectively. For SSD, a heuristic based detecting components with outstanding peaks in the band-of-interest (8–13 Hz) was used, as outlined in the Low-rank factorization section. The resulting method is called SSD AUTO.

in all scenarios. The inability of PCA to recover individual brain oscillations is explained by the fact that PCA, unlike SSD, constrains the activation patterns (the columns of $\mathbf{A}$) to be orthogonal, which is inconsistent with the physical generation of scalp potentials in the EEG. Therefore, PCA is only suitable to recover subspaces of brain signals, not individual signal components.

Number of selected components

The bottom row of Fig. 1 depicts the number of components selected by PCA using the 90% and 99% rules (PCA 90 and PCA 99), as well as of SSD using the heuristic rule presented in the Low-rank factorization section (SSD AUTO). Both PCA 90 and PCA 99 consistently overestimate the number of components of interest, retaining 7 (PCA 90) and 25–26 (PCA 99) components on average regardless of the signal-to-noise ratio and the actual number of simulated oscillators. In contrast, the number of components chosen by SSD AUTO approximates the true number well. Unlike for PCA, this number increases for increasing K_s. For $\lambda = 1/10$ and $K_s = 5$, however, the number of components is slightly underestimated, since, due to the low SNR, fewer components stand out enough to be detected according to Eq. (10).

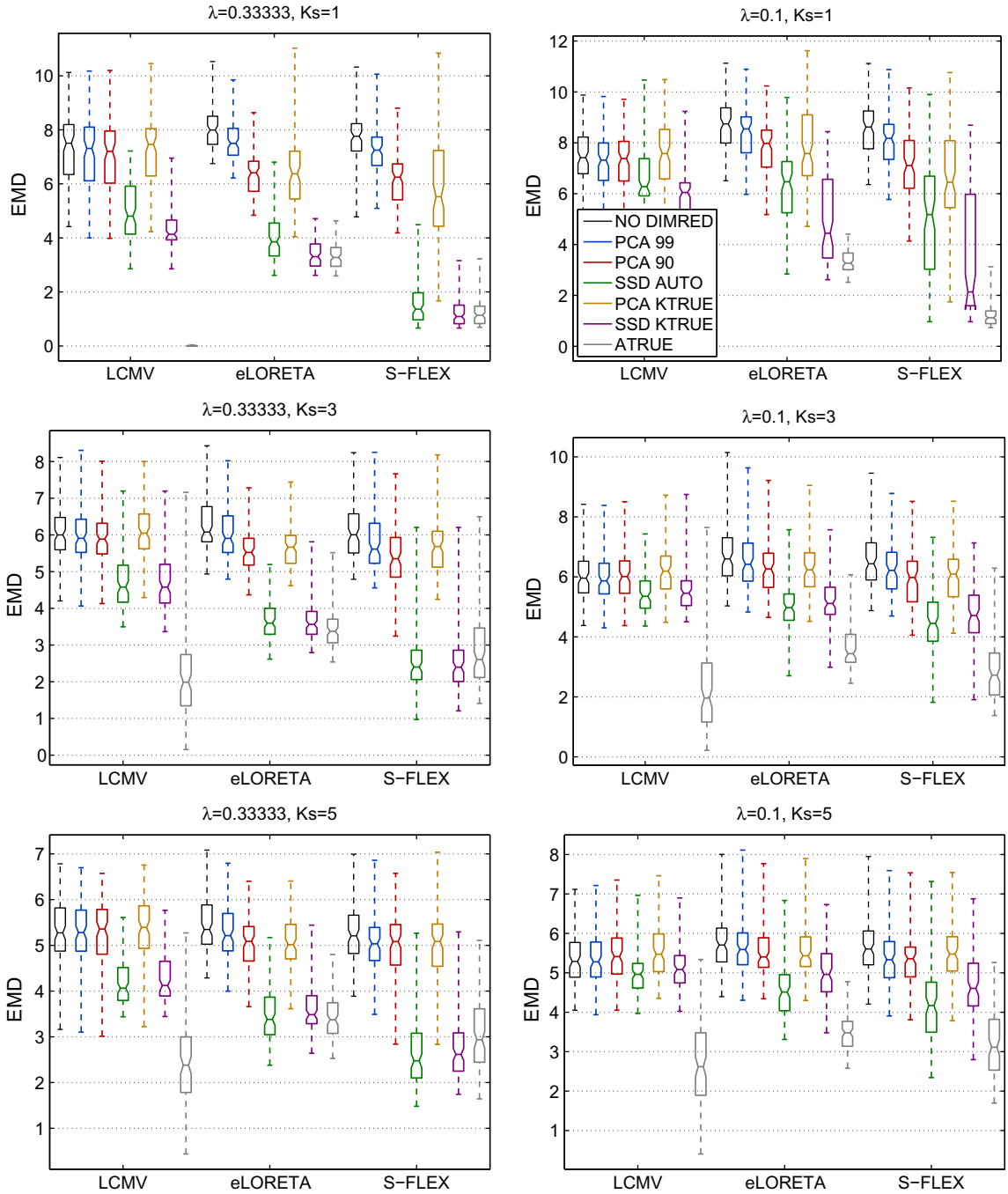

Fig. 2. Quality of EEG inverse source reconstructions of simulated oscillatory activity according to linearly-constrained minimum-variance (LCMV) beamforming, exact low resolution brain electromagnetic tomography (eLORETA) and sparse basis field expansions (S-FLEX) for two signal-to-noise ratios ($\lambda \in \{1/3, 1/10\}$) and three numbers of simulated oscillatory components ($K_s \in \{1, 3, 5\}$). Reconstruction quality is measured in terms of the earth-mover's distance (EMD) between the true and estimated cerebral current power distributions. Shown is the performance obtained using no dimensionality reduction at all (NO DIMRED), using principal component analysis (PCA) in combination with the 90% and 99% heuristics for choosing the number of components (PCA 90 and PCA 99), using spatio-spectral decomposition (SSD) in combination with the heuristics outlined in **the Low-rank factorization section** (SSD AUTO), using PCA and SSD with the true number of simulated oscillations $K = K_s$ (PCA KTRUE and SSD KTRUE), as well as using a projection to the true subspace of the simulated oscillations (ATRUE). Note that PCA KTRUE, SSD KTRUE and ATRUE require information not available in practice and serve only as benchmarks.

Source localization accuracy

Fig. 2 depicts the quality of EEG inverse source reconstructions according to LCMV, eLORETA and S-FLEX when no dimensionality reduction is used (NO DIMRED), when dimensionality reduction according to PCA and SSD is performed (PCA 99,PCA 90, SSD AUTO, PCA KTRUE and SSD KTRUE), and finally when dimensionality reduction to the true subspace of the simulated oscillatory sources is performed (ATRUE). Note that PCA 99, PCA 90 and SSD AUTO represent data-driven heuristics, which may be applied in practice, while PCA KTRUE and SSD KTRUE and A TRUE require knowledge unavailable in practice.

The figure shows a similar performance scheme in all of the six combinations of $K_s \in \{1, 3, 5\}$ and $\lambda \in \{1/3, 1/10\}$ considered. As expected, the lowest performance (largest EMD) is typically achieved for NO DIMRED. In most scenarios, PCA 99 and PCA 90 lead to small improvements, which, however, were not significant. A significant (and in most scenarios substantial) improvement compared to NO DIMRED,

PCA 99 and PCA 90 is achieved with SSD AUTO in all scenarios. Thus, SSD AUTO should be considered in practice when it comes to localizing generators of oscillatory brain activity.

Interestingly, SSD AUTO significantly outperforms the supervised PCA variant PCA KTRUE in all scenarios, indicating that the first K_s PCA components do not at all coincide with the oscillatory subspace of interest, a fact which is also obvious from the top row of Fig. 1. Moreover, the performance using SSD AUTO is typically in the range of the performance of the supervised SSD variant SSD KTRUE, indicating that the heuristic introduced in the Low-rank factorization section is very effective in estimating the number of oscillatory components in the data. In fact, depending on the inverse method used as well as the simulation scenario, either SSD KTRUE significantly outperforms SSD AUTO, SSD AUTO significantly outperforms SSD KTRUE, or both methods are on par. The interesting case where SSD AUTO actually outperforms SSD KTRUE is predominantly observed in scenarios with large K_s and low λ,

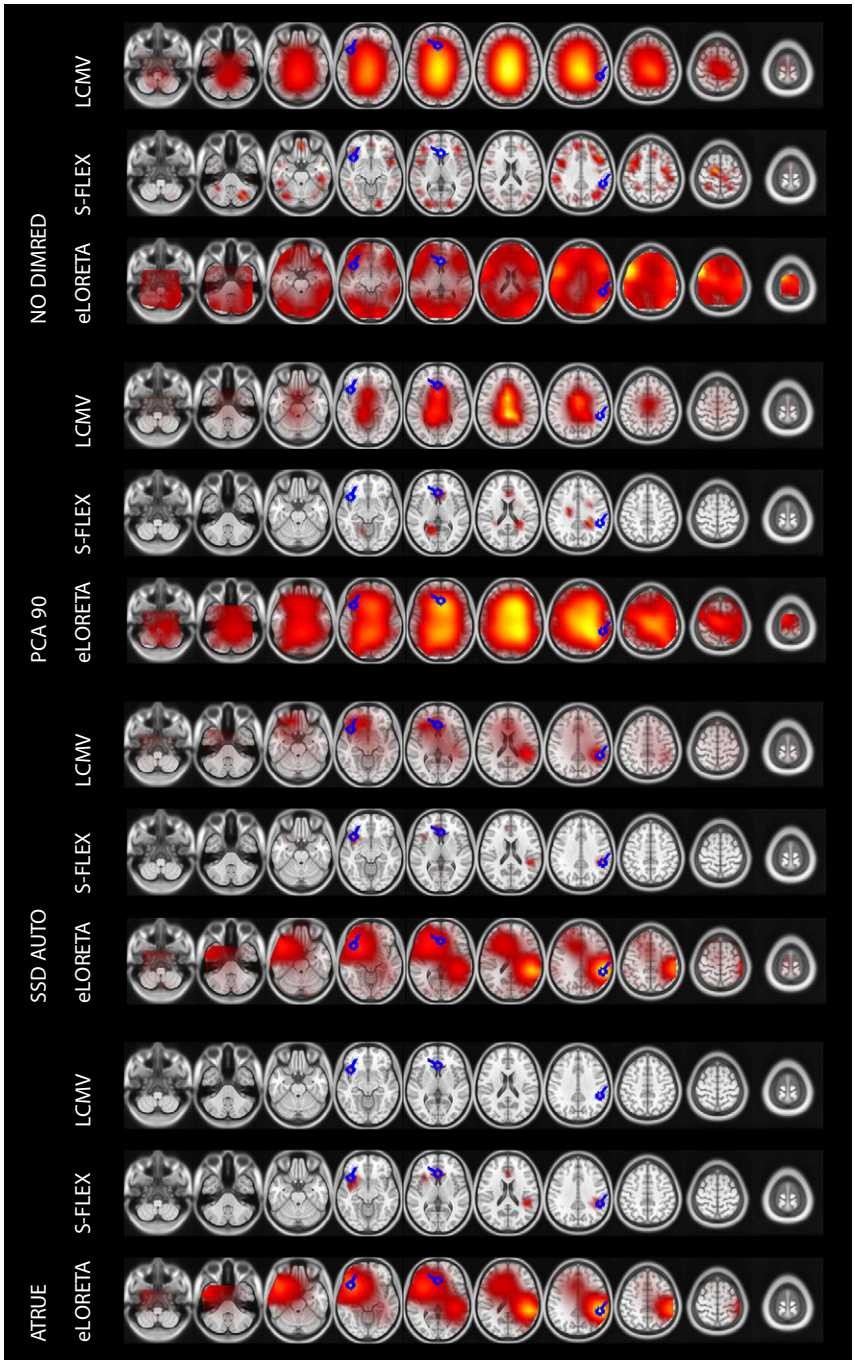

Fig. 3. Typical localization achieved in a single iteration using linearly-constrained minimum-variance (LCMV) beamforming, sparse basis field expansions (S-FLEX), and exact low-resolution tomography (eLORETA) for the scenario $\lambda = 1/3$ and $K_s = 3$. The estimated power in each voxel is depicted as a heat map, where brighter colors indicate higher power. The three simulated electrical dipoles are overlaid as blue arrows. Shown are the localizations using no dimensionality reduction at all (NO DIMRED), using principal component analysis (PCA) in combination with the 90% heuristic for choosing the number of components (PCA 90), using spatio-spectral decomposition (SSD) in combination with the heuristics outlined in the Low-rank factorization section (SSD AUTO), and using a projection to the true subspace of the simulated oscillations (ATRUE). Note that ATRUE requires information not available in practice and serves only as a benchmark.

where SSD AUTO is known to underestimate the true number of oscillatory components (see bottom row of Fig. 1). This indicates that in low-SNR regimes 'later' SSD components, which do not stand out in terms of their spectral peak, are too noisy to be of use for the EEG inverse source reconstruction. In these regimes, it might be more beneficial (in terms of the EMD) to localize fewer oscillators, but with higher accuracy.

Finally, in all but a few specific scenarios, the supervised 'baseline' method A TRUE, which uses knowledge of the true oscillatory subspace, significantly (and often substantially) outperforms all other methods in terms of the EMD, as expected. Exceptions are predominantly observed for high-SNR ($\lambda = 1/3$) in combination with inverse reconstructions according to eLORETA and S-FLEX. In two scenarios ($K_s \geq 3$, S-FLEX), SSD AUTO even significantly outperforms A TRUE. In these cases, the true subspace might actually have a lower SNR as the subspace chosen by SSD's SNR optimization. Performing source localization in the SSD subspace therefore leads to more accurate results at least for this specific inverse method.

Fig. 3 shows a typical localization achieved in a single iteration using LCMV, S-FLEX and eLORETA for the scenario $\lambda = 1/3$ and $K_s = 3$ for NO DIMRED, PCA 90, SSD AUTO and ATRUE. The localization performance using NO DIMRED is generally poor. While the spatial structure of the estimated activity is different for the three methods as a result of their inherently differing assumptions, all have in common that they estimate a significant amount of spurious non-oscillatory activity at a widespread number of locations, while at the same time missing the true activity. For the LCMV beamformer, all estimated activity is concentrated in a large central blob with its center in between the three simulated dipoles. S-FLEX estimates a variety of focal sources scattered across the entire brain, only a part of which are close to the locations of the simulated dipoles. eLORETA estimates activity predominantly at the surface, thereby also mislocating oscillatory source activity. With PCA 90 (similarly with PCA 99 and PCA KTRUE, not shown), the localization quality does not increase. In contrast, the performance of all three methods dramatically improves if SSD AUTO is used. Moreover, their solutions become more similar to each other in terms of spatial characteristics. Centers of estimated activity are observed close to the locations of the three simulated dipoles for all three methods, while activity in other brain areas is largely suppressed. Note that the number of components selected by SSD AUTO in this simulation was correct, i. e., $K = 3$; therefore SSD AUTO and SSD KTRUE are the same method. Notably, the application of SSD leads to results almost indistinguishable from those obtained by projecting to the true oscillatory subspace (ATRUE) if S-FLEX or eLORETA are used. For LCMV, however, knowing the true subspace would lead to a further improvement in localization performance.

Classification of motor imagery conditions

Classification accuracy as a function of the number of PCA/SSD components

Fig. 4 shows the cross-validated classification accuracy of CSP, as well as of PCA + CSP and SSD + CSP, as a function of the number of components used in the dimensionality reduction, separately for the three different electrode montages considered and averaged across 80 subjects. All curves start at six components, because this is the number of CSP components used as features for LDA. Thus six is the minimal number of dimensions of the subspace to which CSP is applied.

The baseline for the comparison is CSP, computed in the original EEG channel space, which is plotted as a straight gray line in the figure. The performance of both PCA + CSP and SSD + CSP first rises as more PCA/SSD components are added until a maximum is reached, even surpassing the classification performance of CSP without dimensionality reduction. As the number of components rises further, the performance of PCA + CSP and SSD + CSP approaches the level of CSP. This is to be expected, because the more components are added, the less dimensionality reduction takes place.

The important difference between the performance curves of PCA + CSP and SSD + CSP is that SSD + CSP reaches its maximum performance much earlier, i.e. with fewer components. The maximum classification performance of PCA + CSP is achieved using a number of PCA components that lies roughly between half and two thirds of the number of EEG channels available. Thus, the optimal number of PCA components depends on the electrode montage used. In contrast, between 10 to 20 SSD components yield the maximum classification performance for SSD + CSP, independently of the number of EEG channels.

Classification accuracy for component selection heuristics

Choosing the number of PCA/SSD components on a subject-specific basis was realized via heuristic selection rules. In Fig. 5 we compare the performance of CSP (no dimensionality reduction) to that of PCA 90 + CSP, PCA 99 + CSP, SSD 14 + CSP, and SSD AUTO + CSP. Note that the choice of including SSD 14 + CSP, which uses a fixed number of 14 SSD components for each subject, was made *after* obtaining the results depicted in Fig. 4. Choosing the first 14 SSD components yielded the (near) maximum classification performance for all electrode montages and may thus serve as an additional benchmark for the heuristic methods.

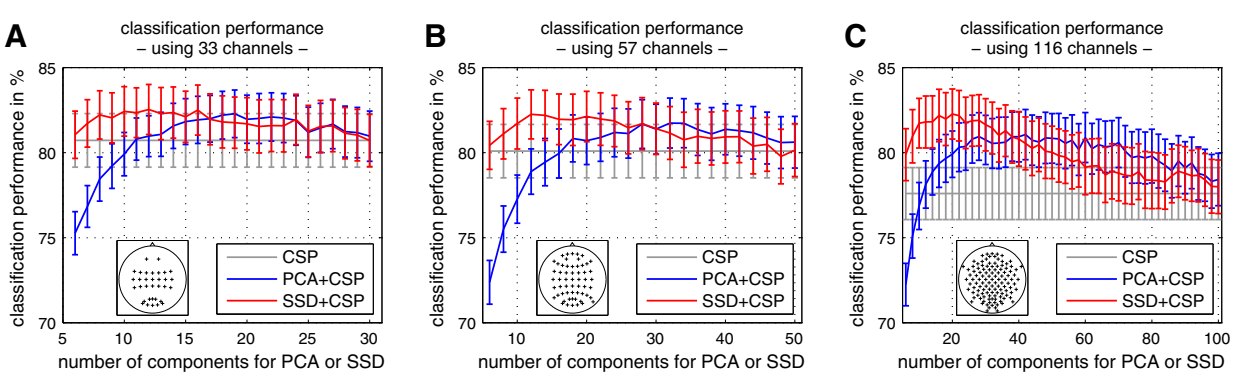

Fig. 4. Cross-validated classification performance in a brain–computer interface setting as a function of the number of PCA or SSD components used for dimensionality reduction of the EEG data, averaged over 80 subjects. The baseline performance (gray line) is given by the application of common spatial patterns (CSP) in the EEG channel input space and subsequent linear classification of CSP log-variance features. Additionally, CSP was performed after PCA (blue line) or after SSD (red line) with a varying number of components being retained. Starting from a set that contained the first six PCA or SSD components, more components were added according to the ordering imposed by their respective (generalized) eigenvalues. The analysis was conducted for three different electrode subsets of the original 119 channel montage, using either (A) 33 channels, (B) 57 channels, or (C) 116 channels. Insets show the respective channel subset layouts. Errorbars indicate standard error of the mean.

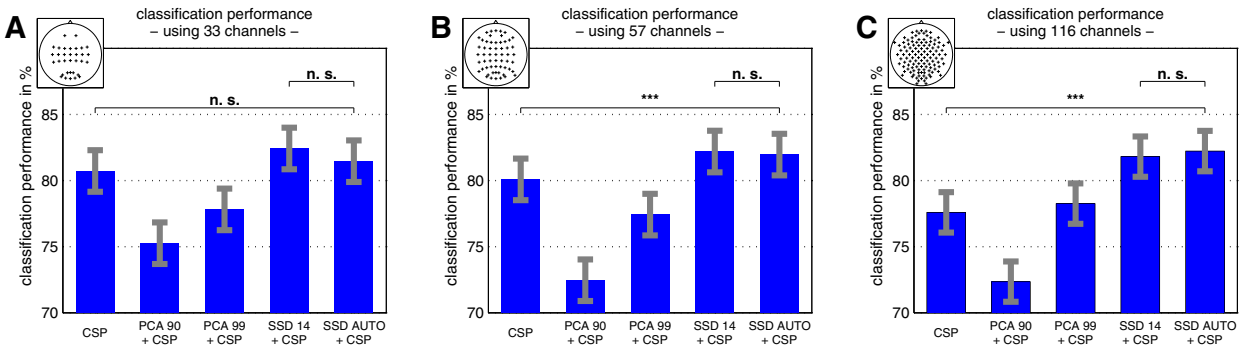

Fig. 5. Cross-validated classification performance in a brain–computer interface setting for different methods of choosing a subject-specific number of PCA or SSD components for dimensionality reduction of the EEG data, averaged over 80 subjects. The baseline performance is given by the application of common spatial patterns (CSP) in the EEG channel input space and subsequent linear classification of CSP log-variance features. Additionally, CSP was performed after the application of PCA or SSD. For PCA, the number of components was set for each subject such that either 90% (99%) percent of total variance were explained, here denoted by PCA 90 + CSP (PCA 99 + CSP). For SSD the number of components was determined according to a heuristic, the resulting method here denoted as SSD AUTO + CSP. Additionally, the performance of using 14 SSD components for each subject is shown here, denoted by SSD 14 + CSP. Note that SSD 14 + CSP is included as a benchmark only, as it yielded the highest performance in the exhaustive sweep across number of components depicted in Fig. 4. The analysis was conducted for three different electrode subsets of the original 119 channel montage, using either (A) 33 channels, (B) 57 channels, or (C) 116 channels. Insets show the respective channel subset layouts. Errorbars indicate standard error of the mean. A single (three) asterisk denote(s) statistically significant differences with $p < 0.05$ ($p < 0.001$), Wilcoxon signed rank test. n.s. not significant.

SSD AUTO + CSP outperformed CSP, PCA 90 + CSP, and PCA 99 + CSP in all electrode montage conditions, the difference between SSD AUTO + CSP and CSP being highly statistically significant for 57 and 116 EEG channels ($p < 0.001$, Wilcoxon signed rank test), while the difference did not reach statistical significance for 33 EEG channels ($p >= 0.05$, Wilcoxon signed rank test).

The performance of SSD 14 + CSP was not affected by the number of EEG electrodes, while the performance SSD AUTO + CSP increased with the number of EEG channels used in the montage. The difference in classification performance between the two methods did not reach statistical significance ($p \geq 0.05$, Wilcoxon signed rank test).

Size of and variance explained by best PCA/SSD subspaces

Fig. 4 shows that an almost maximal performance for SSD + PCA was achieved already with the initial six components, and that the later addition of SSD components only slightly increased the classification performance. In this sense an interesting question is the following: with how few components (PCA or SSD) does the classification performance reach 95% of its maximal value? The remaining increase is rather small and its very gradual behavior may require many more components, without in fact contributing substantially to the classification. Histograms showing the distribution of this number of components across subjects are shown in Fig. 6 for 100% and 95% of the individual maximal performance for the 57 channel montage. It can be seen that,

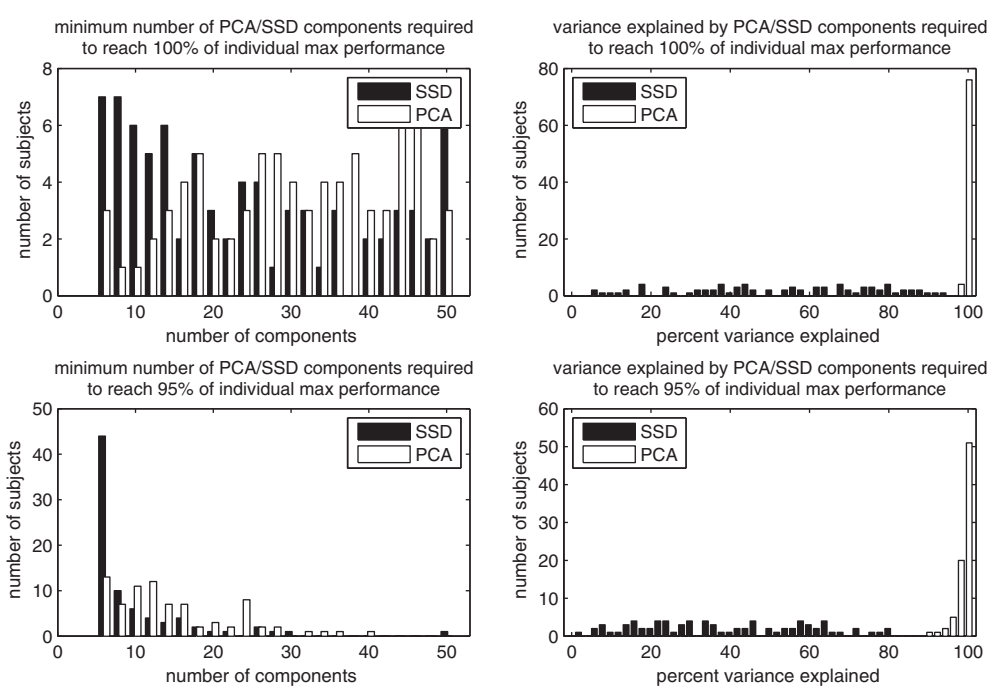

Fig. 6. *Left column*: Histogram across 80 subjects of the number of SSD or PCA components required to reach 100% (top row) or 95% (bottom row) of subject-specific maximal classification performance in the brain–computer interface setting (57 EEG channels, CSP and LDA applied to dimensionality reduced EEG data). *Right column*: Histogram of the variance explained by the SSD or PCA components required to reach 100% (top row) or 95% (bottom row) of subject-specific maximal classification performance.

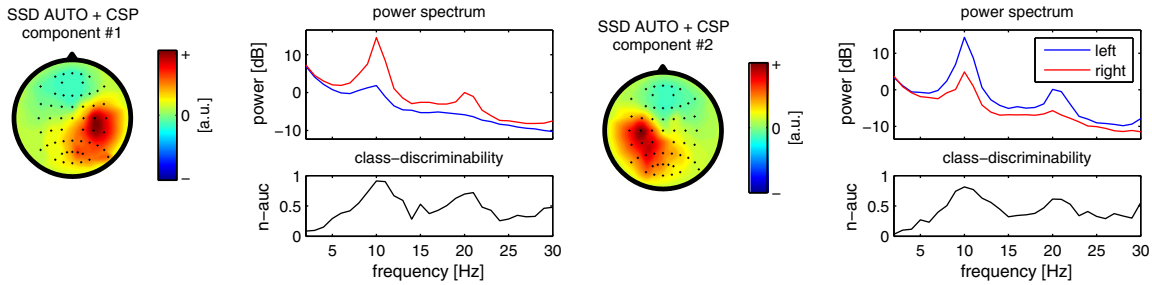

Fig. 7. Spatial activation patterns and class-wise power spectra of two CSP components obtained after dimensionality reduction with SSD AUTO. Displayed are the two most class-discriminative components, with the most discriminative to the left. Class-discriminability was assessed by normalized area under the ROC curve (n-auc) for the log-variance of the subject-specific frequency band. n-auc is defined to be between 0 and 1. The power spectrum shows two lines, one for trials in which movement of the left hand was imagined (blue line) and one for the trials in which movement of the right hand was imagined (red line). Below the power spectra, the corresponding class-discriminability in Fourier space is shown, i.e. the single-trial separability of the above power spectra curves, measured by the normalized area under the ROC curve.

in order to reach 95% percent of maximal performance, 10 or fewer SSD components are sufficient for the vast majority of subjects, i. e., more than 60 out of 80 subjects. Using PCA for pre-processing, for the vast majority of subjects ten or more components are necessary to reach 95% of individual maximum performance.

Fig. 6 also shows the cumulative variance explained by the PCA/SSD components required to reach the given levels of performance. 'Cumulative variance explained' signifies how much of the total variance contained in the data can be accounted for by a given subspace, here the subspace of components required to reach a certain level of performance. The histograms of variance explained are remarkably different for PCA and SSD subspaces. While the distribution of variance explained for SSD components is very broad, has no clear peak, and is mostly concentrated in the less-than 90% range of total variance, the distribution for PCA components is almost exclusively contained in the more-than 90% range of total variance.

A representative subject

Fig. 7 provides a typical example of spatial topographies obtained after the dimensionality reduction with SSD. It shows the two best class-discriminating components for a representative subject obtained with SSD AUTO + CSP. For each component we show its spatial activation pattern, the class-specific power spectrum as well as normalized area under the receiver-operating characteristics curve (n-auc) values for each frequency bin, which indicate the class-separability for each frequency bin.

For this subject, the selected motor-imagery classes were left hand and right hand. Accordingly, the spatial pattern of the best discriminating component is consistent with a dipolar source, located in the right sensori-motor region of the cortex (showing an ERD for trials of the 'left' class). The power spectrum of that component shows a strong desynchronization of the sensori-motor rhythm (SMR) in the alpha range (as well as its first harmonic in the beta range). The second-best class-discriminating component is consistent with a dipolar source in the left motor cortex and exhibits a differential spectral response that is opposite to that of the first component.

Discussion

The results of our study unequivocally show that SSD can serve as a reliable pre-processing step for dimensionality reduction in data sets containing oscillatory components. This was confirmed for simulated and real EEG data. Importantly, SSD pre-processing is not restricted to EEG recordings. Multichannel recordings of neuronal activity are also performed with MEG, ECoG and LFP. All these techniques are capable of capturing neuronal oscillations but are also affected by volume conduction and represent a superposition of signal of interest and noise. Therefore, a reduction of the number of dimensions as well as a separation of signal from noise for further analysis is as imperative for these recoding techniques as for EEG.

Our results are particularly relevant for cases in which multivariate pattern analysis (MVPA) is to be conducted with respect to a particular feature of neural oscillations. In our example on BCIs, dimensionality reduction with SSD increased the performance of the MVPA method CSP, which extracts sources with maximal difference in spectral power between two conditions. We expect this positive effect to carry over to cases in which the subjects of investigation are, for example, ongoing source envelope dynamics (Blythe et al., 2014; Dähne et al., 2013; Dähne et al., 2014a; Dähne et al., 2014b) or connectivity between neural sources (Ewald et al., 2012; Gómez-Herrero et al., 2008; Haufe et al., 2010; Nolte et al., 2006).

Application of SSD to multichannel recordings

Similar to popular ICA algorithms, SSD assumes mutually decorrelated sources and, indeed, the SSD solutions (i.e. the generalized SSD eigenvectors) are constrained such that the projected signals are mutually decorrelated. For a discussion of the conditions required for successful SSD decompositions please see (Nikulin et al., 2011). Importantly, even if SSD, when being applied as a pre-processing step, does not discriminate completely between individual oscillatory sources, it still significantly reduces the variance of the data and maximizes SNR. A further decomposition of the sources can be achieved with subsequently applied inverse modeling techniques. In this sense SSD is not a substitution to inverse modeling but rather a complementary procedure ensuring a proper extraction of the neuronal activity to be localized.

Importance should also be given to the determination of the flanking frequencies for SSD. The size of the flanking frequency bands depends on the frequency range at which SSD is applied. For relatively narrow alpha oscillations, flanking bands of about 1–2 Hz width can be used. Greatly enlarging frequency bands might lead to a situation when flanking frequencies would also include harmonics of the base frequencies; for instance 20 Hz oscillations. Since alpha and beta oscillations might not represent two different oscillatory phenomena but rather complimentary descriptions of non-sinusoidal oscillations (Gaarder and Speck, 1967; Nikulin and Brismar, 2006), inclusion of beta oscillations in the flanking frequencies might result in the situation that the central and flanking bands would include the same neuronal source and the optimization for SNR might be problematic. However, in case of high frequency oscillations, such as sigma oscillations (~ 600 Hz, Fedele et al., 2012), the peaks are very broad with widths of hundreds of Hz and thus flanking frequencies can be as broad as tens of Hz.

SSD versus PCA

The main reason why SSD outperformed PCA in our study is the fact that SSD decomposes the data not on the basis of variance but rather by selecting components on the basis of how much the spectral peak stands above the neighboring, flanking, frequencies, which, as was shown earlier, is equivalent to maximizing the signal-to-noise ratio (Nikulin et al., 2011). Commonly, EEG and MEG spectra are dominated by 1/f noise, which constitutes the major component of the neuronal noise in electrophysiological recordings. For many oscillations, including even alpha oscillations (particularly sensorimotor alpha oscillations, a. k. a. the mu rhythm), the 1/f part of the spectrum is so strong that the peaks of the oscillations are not even visible. This in turn implies that, when performing PCA, a large number of components might contain not only oscillations but to a large extent also 1/f noise. Clearly, bringing noise to further calculations would only decrease the efficacy of the main analysis, be it source modeling or single trial estimation. SSD, however, does not optimize for signals with large variance, but rather selects signals with large SNR. This, on one hand, leads to the smaller number of components selected for further analysis (see Fig. 1 for source modeling and Fig. 6 for the classification accuracy). On the other hand SSD includes only a relatively small fraction of variance of the whole EEG data, compared to 90–99% of the total variance used in case of PCA (Fig. 6). Reduction in the number of components and exclusion of noise consequently results in better performance w. r. t. inverse modeling and BCI classification.

SSD versus ICA

Another alternative to PCA is independent component analysis (ICA, Hyvärinen et al., 2004; Makeig et al., 2004). ICA is effective for removal of electrooculogram or muscle activity contaminating EEG/MEG data. This is because these noise sources are characterized by different higher order statistics (e. g. kurtosis, skewness) compared to genuine neuronal activity, and ICA techniques are often based on differentiation of components according to these measures. The situation with oscillations is different, and standard ICA techniques are not as efficient in differentiation of oscillatory sources (Hyvärinen et al., 2010; Nikulin et al., 2011).

The intention of the present study was not to benchmark SSD against ICA techniques. In fact this has already been done in the previous study (Nikulin et al., 2011), where we have shown that SSD outperforms ICA based on FastICA (Hyvärinen and Oja, 1997), Infomax (Bell and Sejnowski, 1995; Delorme and Makeig, 2004) and SOBI (Belouchrani et al., 1997) in terms of extracting oscillatory sources. We rather aimed here at showing that SSD is vastly superior to the routinely used PCA, despite the fact that computationally SSD is quite similar to PCA as generalized eigenvalue decompositions can be computed in a few milliseconds. The speed of calculation is an important factor, since the pre-processing stage per definition should not take a long time compared to the following main analysis. Another advantage of SSD over ICA is that one does not need to perform a manual inspection of the obtained SSD components; they are rather selected on the basis of their eigenvalues similar to PCA. In case of ICA, one often needs to visually inspect spectra, time courses of the components or their topographies.

Dimensionality reduction versus low-rank factorization

As outlined in the Dimensionality reduction and Low-rank factorization sections, there are two general ways in which linear data decompositions like PCA and SSD can be used to denoise data: dimensionality reduction and low-rank factorization. Here we used both approaches in different contexts. In the classification setting (see subsection Classification of motor imagery conditions under Methods and Classification of motor imagery conditions under Results section), the denoised data are used as input for further feature extraction and classification routines. These further processings can greatly benefit from input data with fewer dimensions as provided by Eq. (1). In the source localization setting (see subsection Localization of simulated oscillatory EEG sources under Methods and Localization of simulated oscillatory EEG sources under Results section), on the other hand, the 'denoised' data are approximated by brain sources according to the generative model Eq. (11) for the purpose of source localization. This model holds for EEG data in original sensor space; therefore, any dimensionality reducing transformation applied to the data according to Eq. (1) on the left hand side also has to be applied to the lead field $\mathbf{L}$ on the right hand side. However, doing so results in a general loss of information, which, depending on the method used, may degrade source localization performance. Precisely, the transformed lead field $\mathbf{W}^\top\mathbf{L}$ in the reduced system $\widetilde{\mathbf{x}}(t) = \mathbf{W}^\top\mathbf{x}(t) = \mathbf{W}^\top(\mathbf{L}\mathbf{j}(t) + \epsilon(t))$ does not contain any information about brain sources contributing to the *null space* of $\mathbf{W}^\top$, which is exactly the space spanned by the noise, anymore. This has the consequence that inverse methods based on minimizing a likelihood term may place brain sources at arbitrary locations contributing to this noise space without changing the likelihood of the transformed data $\widetilde{\mathbf{x}}(t)$. This is avoided by working in the original space using the low-rank factorization $\widetilde{\widetilde{\mathbf{x}}}(t)$ of $\mathbf{x}(t)$ in conjunction with the original lead field $\mathbf{L}$. In the modified system $\widetilde{\widetilde{\mathbf{x}}}(t) = \mathbf{L}\mathbf{j}(t) + \epsilon(t)$, all information about electrical sources contributing to the noise space is still contained in the lead field $\mathbf{L}$. The absence of noise activity in $\widetilde{\widetilde{\mathbf{x}}}(t)$ therefore effectively suppresses activity in brain areas contributing to this noise space, with favorable consequences for source localization accuracy.

Regularization

Although different regularizations can also be considered for the data after the application of PCA and SSD, their discussion goes beyond the scope of this study. Such regularizations rather relate to each specific method (inverse modeling, BCI etc). The main idea of the preprocessing, as presented here, is rather to introduce the procedure which minimizes the necessity for additional regularizations.

Source separation in the presence of 1/f noise

Nikulin et al. (2011) showed that not only SSD is capable to reliably separate oscillatory sources from neuronal 1/f noise in simulations and real data but that even individual sources could be separated better than with ICA techniques. This result is in agreement with the result of the present study, where the angles between SSD-related subspaces and original source subspaces were much smaller compared to the subspaces obtained with PCA. Such accurate SSD extraction of original oscillatory sources is particularly important for the following modeling of non-averaged oscillatory dynamics. Extraction of ongoing non-segmented oscillatory time-courses is for instance relevant for studies related to the investigation of the criticality hypothesis (Linkenkaer-Hansen et al., 2001), where some recent studies reported differential neuronal dynamics depending on specific brain areas (Palva et al., 2013), and also for studies investigating large-scale cortical correlation structure (de Pasquale et al., 2012; Hipp et al., 2012). For such a fine-grained differentiation of the origins of neuronal oscillations it is important to take into account that even ongoing alpha oscillations have only moderate to poor signal to noise ratio, and an application of inverse modeling to such data without proper pre-processing might be quite inaccurate as follows from the results of the present study. The situation is even more aggravated when one wants to extract temporal dynamics of faster oscillations in the beta and gamma ranges, since these oscillations have even smaller amplitude and are often not even distinguishable above the 1/f part of the spectrum.

It is also the case that mass-univariate statistics are used for EEG and MEG analysis in sensor space and source space for instance after the application of beamformer techniques. The use of SSD pre-processing can also be beneficial in this case for the following reasons. SSD acts as

a denoising technique, effectively removing 1/f noise in the frequency range of interest. As any inverse decomposition technique is sensitive to the SNR, SSD noise-removal is beneficial for the following source reconstruction. Moreover, in many cases SSD by itself is capable to differentiate between different oscillatory sources within the same frequency range to such an extent that inverse solutions become easier where individual components can be modeled even with single equivalent current dipoles thus eliminating the necessity to perform adjustments for multiple comparisons.

Implications for neuroscience and clinical studies

An accurate source reconstruction is particularly critical for clinical studies, where the identification of pathological cortical areas is important. For instance, ECoG study by Monto et al. (2007) showed that interictal temporal dynamics of beta oscillations are particularly affected in the spatial proximity of the epileptogenic foci and thus an accurate spatial reconstruction of beta generators is important. Other examples, where a proper extraction of ongoing neuronal oscillations is important in clinical studies include Parkinson's disease (Hohlefeld et al., 2012), Alzheimer disease (Montez et al., 2009) and schizophrenia (Nikulin et al., 2012). Therefore, since in the present study SSD was capable of significantly improving a localization of generators of brain activity, we believe that a pre-processing with SSD can facilitate the identification of neuronal sources characterized by pathological oscillatory dynamics.

In our study, single trial analysis was performed in the context of BCI. Although BCI research as such aims at providing subjects with the ability to control peripheral devices or computer programs with changes in the EEG signal, the main BCI processing steps such as extraction of relevant features and the following classification represent essential steps for any single trial analysis, be it later used in a BCI context or in general neuroscientific experiments on high cognitive functions (Pernet et al., 2011). Therefore, an increase in classification accuracy in motor imagery BCI after the application of SSD can in general be considered as a proof of principle for the supporting role of SSD in single trial analysis.

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